Privacy Vulnerabilities in NYC Taxi Trip Data

Project Discussion

Group 16

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Motivation

The Open Data Dilemma:

- Cities publish transportation data for transparency and research
- Assumption: Removing names = Privacy protected
- Reality: "Anonymous" data can be re-identified

Real-World Impact:

- 2014: NYC taxi data exposed celebrity movements, mosque visitors
- 2019: Researchers linked "anonymized" location data to individuals
- Our Goal: Quantify how easily re-identification occurs & recommend some changes to improve data

Research Question:

"How vulnerable is published taxi data to privacy attacks, and can we measure the privacy risk?"

Problem Statement

Dataset:

- NYC Yellow Taxi Trip Records (3.5M+ trips, Jan 2019 to Aug 2025)
- Published openly with "identifying information removed"
- Contains: locations, timestamps, fares, tips, payment methods

The Challenge is combinations of seemingly harmless attributes can uniquely identify individuals

Privacy Threats:

1. Background Knowledge Attacks: Minimal info → Full trip details
2. Pattern Inference: Reveal home/work locations, habits
3. Sensitive Data Exposure: Financial behavior, lifestyle patterns

Our Mission: Prove the vulnerability exists and measure its severity

Methodology

Exploratory Data Analysis (EDA)

Privacy Metrics Evaluations

- k-Anonymity Testing
- L-Diversity Analysis
- T-Closeness Analysis

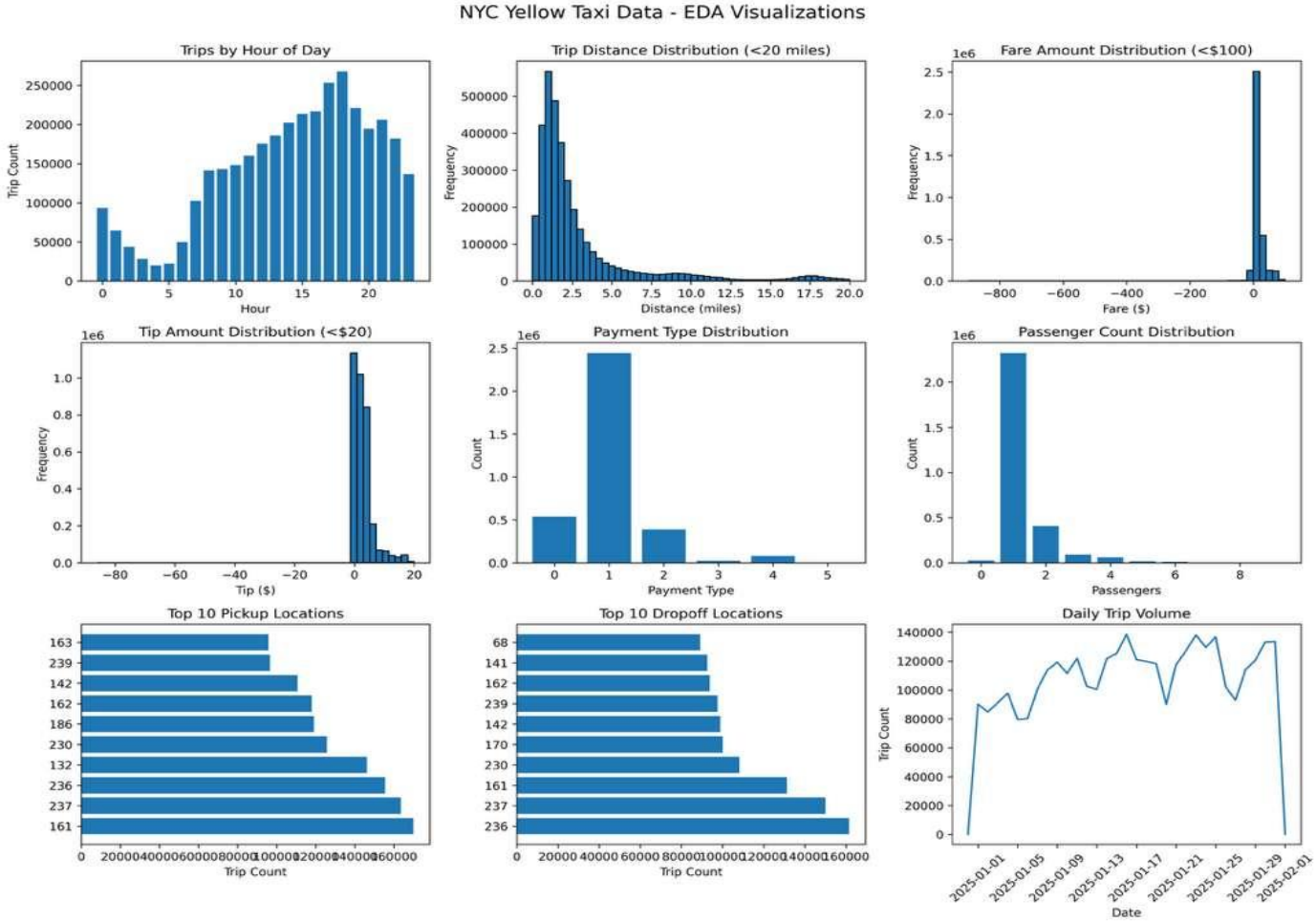
Attack simulation

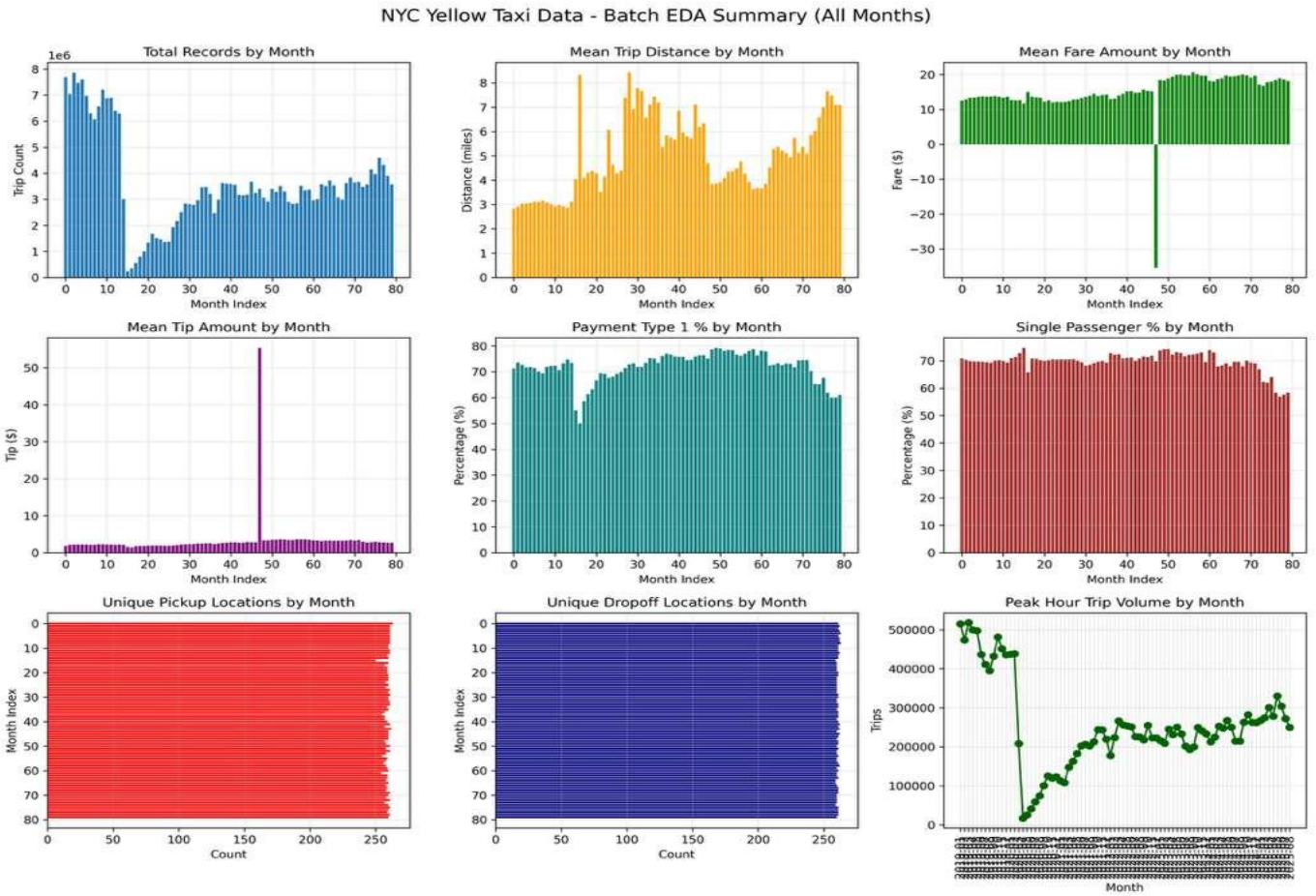
- Socioeconomic Profiling Attack
- Payment Behavior Deanonymization Attack
- Travel Pattern Inference Attack

Privacy v/s Utility Analysis

- Generalization Lattice
- Testing on a ML task

Exploratory Data Analysis


NYC Yellow Taxi Data - EDA Visualizations
Trips by Hour of Day
Trip Count
Hour
Trip Distance Distribution (<20 miles)
Frequency
Distance (miles)
Fare Amount Distribution (<$100)
1e6
Frequency
Fare ($)
Tip Amount Distribution (<$20)
1e6
Frequency
Tip ($)
Payment Type Distribution
1e6
Count
Payment Type
Passenger Count Distribution
1e6
Count
Passengers
Top 10 Pickup Locations
163
239
142
162
186
230
132
236
237
161
Trip Count
Top 10 Dropoff Locations
68
141
162
239
142
170
230
161
237
236
Trip Count
Daily Trip Volume
Trip Count
Date


NYC Yellow Taxi Data - Batch EDA Summary (All Months)
Total Records by Month
1e6
Trip Count
Month Index
Mean Trip Distance by Month
Distance (miles)
Month Index
Mean Fare Amount by Month
Fare ($)
Month Index
Mean Tip Amount by Month
Tip ($)
Month Index
Payment Type 1 % by Month
Percentage (%)
Month Index
Single Passenger % by Month
Percentage (%)
Month Index
Unique Pickup Locations by Month
Month Index
Count
Unique Dropoff Locations by Month
Month Index
Count
Peak Hour Trip Volume by Month
Trips
Month

Comprehensive K-Anonymity Analysis

Scenario	Average Unique k=1	Average violation k<5	Worst Period
Location Only	27.71%	54.68%	2020-04 (37%)
Location + Time	39.28%	63.66%	2020-04 (53%)
Location + Time + Distance	43.10%	68.13%	2020-04 (57%)
Location + Time + Passenger	**45.62%**	**73.52%**	2020-05 (52%)

Problems

CRITICAL FINDINGS:

- ALL 80 months show k=1 - complete privacy failure across 6+ years
- COVID period (2020): 96.3% trip reduction → 53% unique identifiability (WORST)
- Privacy worsening: 2025 shows 42.65% unique vs 35.90% in 2019 (19% increase)
- Never safe: Even highest volume months (7M+ trips) show 32%+ unique records

Next Steps

1. **Generalization Lattice & K-Anonymity**
    a. Create generalization lattice for quasi-identifiers
    b. Analyze k-anonymity at each node using brute force

2. **Attack Scenarios**
    a. High vs. low tipping areas (driver exploitation)
    b. Cash vs. card payment patterns by location
    c. Airport travel times (identifying empty homes)

Next Steps

3. **Privacy vs. Utility Trade-off**
   a. ML Task: Train a model to predict tip amount based on pickup location, time, trip distance, and fare
   b. Compare model performance (accuracy, RMSE) on original vs. anonymized data
   c. Evaluate how different anonymization levels impact prediction quality

Questions

1. Any other task that we can use for privacy vs utility comparison?
2. Should we include more data?
3. Any other task or analysis to be added?
4. Should we calculate l-diversity or t-closeness once we achieve a certain level of k-anonymity?